Modelling of Launch Vehicle Radiated Emission at Payload Interface during Fairing Separation

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***Abstract*— This paper describes the development of a simulation model to estimate the ElectroMagnetic Compatibility between a launch vehicle and its payload. This was needed to estimate the E-field radiated towards a payload surface by the launcher antennas during the fairing separation transient.**

The model was developed in the ANSYS HFSS SBR+ simulation environment, and it included CADs of the structure, dynamics of fairing separation and measurements of the LV antenna radiation patterns. The resulting model allowed to simulate various instants of the separation sequence, showing an E-field increase with a peak at around 0.1 seconds after separation. Then, the E-field quickly decreased until, after about one second, the contribution of the fairing reflection became negligible.

Keywords—EMC Model, Launch Vehicle, Fairing Reflection, ANSYS HFSS

I. INTRODUCTION

When determining the compatibility between a Launch Vehicle (LV) and payload satellites, it is critical to verify that electromagnetic compatibility between all parts is guaranteed. This process consists of two main checks: on one hand it is necessary to verify that the payloads will not interfere with the LV functionalities, and on the other hand it is necessary to ensure that the LV emissions (spurious and intentional) will not damage the payloads. Often, these checks are trivial thanks to a series of precautions and considerations, such as satellites having most of their RF instrumentation turned off during launch.

Nevertheless, sometimes there are exceptions that need to be analysed in detail. Such was the case for a candidate payload which could have been susceptible to very short transients of high E-field. For this constraint, the worst-case scenario during launch was identified with the fairing separation transient. In fact, when the fairing separates, it briefly falls in front of the LV antennas and, since the fairing is made of a mostly reflective material, it could reflect part of the signal from the LV antennas back towards the payload.

However, as the fairing separates during a phase in which the LV is accelerating, and thus it very quickly falls away from the launcher (order of few hundreds of milliseconds) as shown in Fig. 1, this phase was usually not analyzed in detail for what regards EMC.

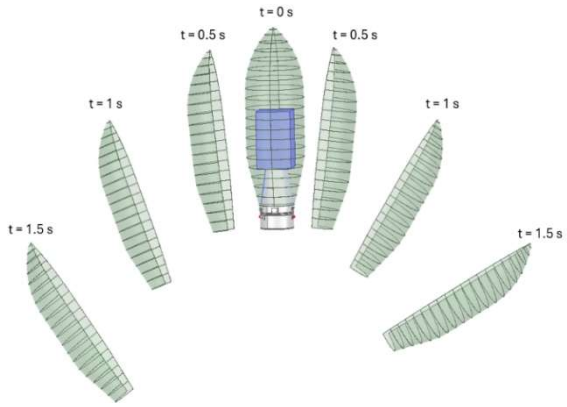


Fig. 1 Position of the two fairing shells at various instants during the fairing separation sequence

Similarly, efforts in the literature have typically focused on studying the E-field within a closed fairing [1][2][3] or the effect of RF attenuators placed inside the fairing [4][5], but no information is readily available regarding the interaction of the fairing with the LV antennas (which is also strictly dependent on the launcher configuration and geometry).

Furthermore, given the dimensions of the fairing (each shell is about eight meters tall), a verification by test would have been extremely complicated in terms of facilities, logistics and costs. The best option was therefore to develop an EMC model and simulate the transient to estimate the maximum E-field at the payload interface. It was decided to create this model using ANSYS HFSS SBR+ environment [6].

This is an enhanced version (proprietary of ANSYS) of the Shooting-and-Bouncing Ray (SBR) solver method [7], a high-frequency ray-tracing approximation that efficiently computes scattering and can be thought of as a hybridization of geometrical optics (GO) and physical optics (PO). The SBR methodology is intended to predict scattering from large-scale geometries, such as an antenna installation environment or platform, and therefore is suited for the scope of this analysis.

The choice of this solver was also driven by considerations regarding hardware requirements and the need to quickly iterate over multiple simulations: in fact, methods that could be more accurate, like the Method of Moments (MoM) or the Electric Field Integral Equations (EFIE), would require a very high computational cost, needing dedicated clusters or cloud computing to solve in reasonable times for such a large geometry. Instead, the model using SBR+ was executed on a computer with 64 GB of RAM and an integrated intel GPU, with simulation times in the order of an hour per iteration.

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II. MODEL DESCRIPTION

The first step in the creation of the model was to import into ANSYS HFSS the CAD elements of interest. These are shown in Fig. 2 and include:

- A simplified model of the LV upper stage, which is where the LV antennas are mounted.
- The payload adapter (PLA). In this case it was a simple section of a cone, but different missions might use different PLAs.
- A volume envelope of the payload under investigation (here generalized to a 2x2x3 meters box envelope), with two sides parallel to the fairing separation axis.
- The two shells of the payload fairing.

In this model, all the geometries mentioned above were implemented as Perfect Electrical Conductors (PEC). This assumption bypassed the complications linked to modelling each material while still being realistic, as most LV structures are made of conductive materials.

Once the geometries were imported, the position of each fairing shell was parametrized according to its foreseen trajectory with respect to the LV during the separation transient, allowing to study the time evolution of the E-field on the payload envelope in discrete steps.

The following step in the model creation was the implementation of the LV antennas, which were the E-field sources for these simulations. Two antennas were considered, placed at diametrally opposite positions on the LV upper stage facing the fairing shells, as shown in Fig. 2.

As the radiation pattern of the LV antennas was recently measured during a test campaign in a spherical near-field measurement facility, it was possible to directly import into ANSYS HFSS the measured data without having to model the antenna itself.

However, one intermediate step was necessary: in fact, as the measured radiation pattern was imported into ANSYS

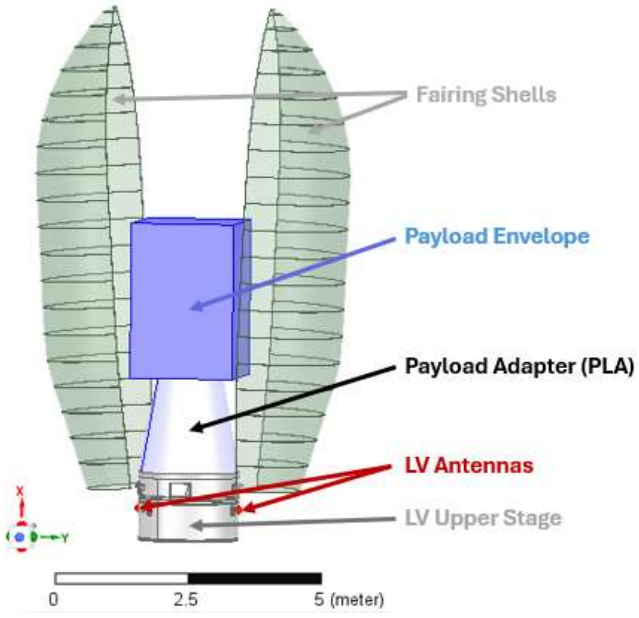


Fig. 2 CAD elements of the model, including the fairing shells, an envelope of the payload volume, the LV payload adapter and the LV upper stage. The position of the two LV antennas is also shown (red dots)

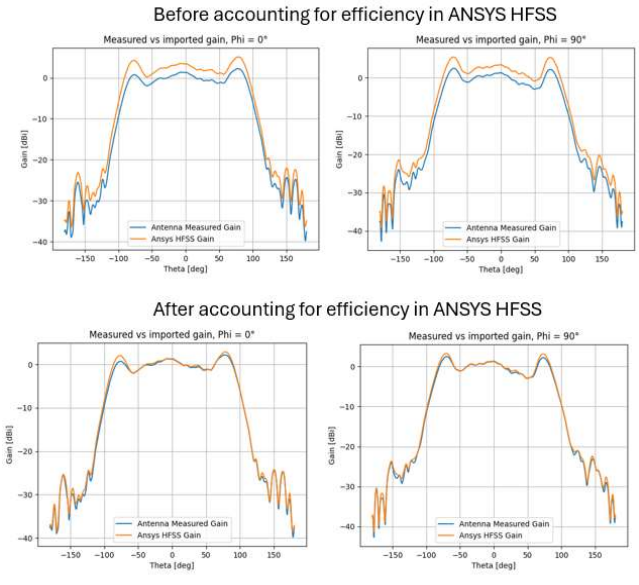


Fig. 3 Comparison of measured (blue) and simulated (orange) gain of LV antenna before and after adding a scaling factor in ANSYS

HFSS without an antenna structure; the model had no information regarding the antenna efficiency (including ohmic and reflection losses), and therefore the simulated directivity and gain values were at first considered identical. However, the test performed also measured the antenna gain, which was therefore used to estimate its efficiency. This allowed to add a scaling factor in the ANSYS HFSS model to ensure that the simulated antenna gain matched the measured antenna gain.

The value of the scaling factor was chosen to be -2.2 dB, which corresponds to an efficiency of the antenna of 0.6, which was consistent with the known characteristics of the antenna. As shown in Fig. 3, with this scaling, the simulated gain overlapped well the measured antenna gain (with small deviations, but it was verified that the simulated gain was always higher than the measurement as a form of additional margin).

III. MODEL VERIFICATION

To verify that the imported antenna behaved as expected within the model, a sanity check was performed by considering an antenna supplied with a 1 W input and mounted on a PEC cylinder (representing the LV upper stage), and simulating the E-field directly in front of the antenna at a 1 meter distance, as shown in Fig. 4.

This setup was particularly useful as it was comparable to a calibration measurement that was performed in an anechoic chamber in the frame of a separate LV EMC test, where the E-field generated by the LV antenna mounted on the LV upper stage was measured by a receiving horn antenna placed at a one-meter distance.

Finally, as a further check, Friis formula (1) was also used to estimate the E-field radiated at a distance d = 1 meter from the LV antenna, considering an antenna gain G_t ≈1.3 dBi in the boresight direction and an input power to the antenna P_t = 1 Watt

$$E_{dB\mu V/m} = 10\log_{10}(Z_0/4\pi) + 10\log_{10}(P_t + G_t) - 20\log_{10}(d) + 120. \quad (1)$$

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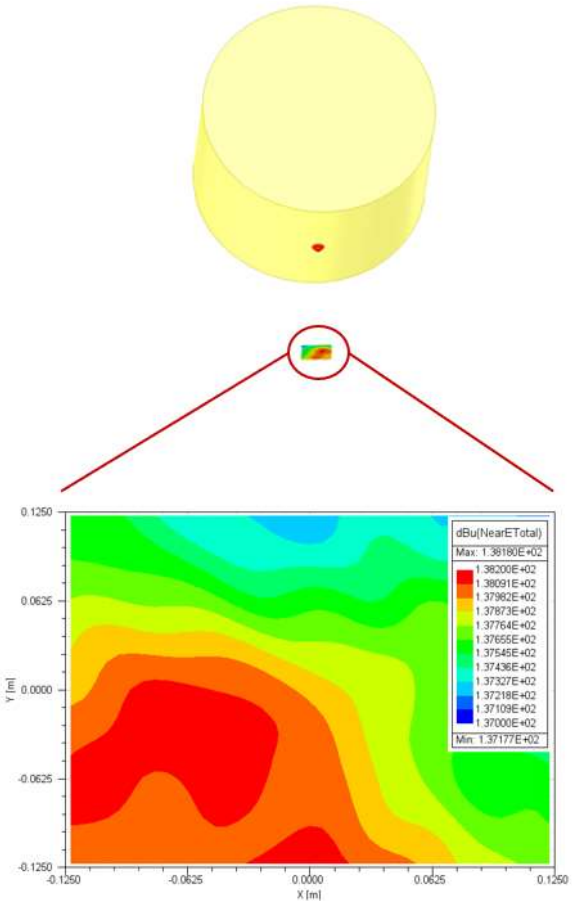


Fig. 4 Simulation setup with control surface 1 meter in front of the antenna placed the LV upper stage (simplified as a cylinder). The simulation surface was chosen to have the same dimensions as the horn antenna used for the actual test.

The comparison between the three results is reported in Table 1 and showed that the simulation overestimated the E-field by almost 3 dB with respect to the measurement. This difference was likely due to a combination of assumptions and approximations, such as:

- The radiation pattern of different serial numbers of the same antenna model might differ slightly due to manufacturing.
- Surfaces were considered smooth and PEC, which could lead to slightly higher reflection from the LV upper stage than in the real system.
- Near-field interactions of the antenna (not accounted for as the imported pattern represents far-field)
- SBR+ solver approximations

These are all points of improvement for future developments but, as the model was overestimating, they were accepted as an additional form of margin leading to more conservative results.

Table 1 E-field comparison between measurements and simulated results at a distance of one meter from the antenna on the LV upper stage

E-field Measured in test [dBµV/m]	E-field calculated using (1) [dBµV/m]	E-field Simulated in ANSYS [dBµV/m]
135.3	136.0	138.2

IV. RESULTS

After having verified that the model results were sensible, the E-field was simulated on an array of points on the payload envelope surface at instants [0.05; 0.1; 0.2; 0.3; 0.5; 1.0; 1.5] seconds after fairing separation, considering two emitting antennas with a steady input power of 1 W each.

The results are summarized in Fig. 6, which reports the distribution of the E-field on the payload surface. As expected, the model showed that immediately after fairing separation the E-field levels increased as the payload was exposed to both the direct illumination from the antennas and the reflection on the fairing shells, with a peak being reached around 0.1 seconds after fairing separation. As the fairing shells started to get further from the LV, the E-field levels gradually decreased, and after about 1 second the effects of the fairing reflection were already almost negligible.

Furthermore, Fig. 7 shows a representation of the separation sequence in which the spatial distribution of the field can be observed. It can be seen that, as expected, the side of the envelope that was directly facing the fairing shell was the most affected by the E-field increase.

Finally, an additional feature of the model allowed to decompose the E-field in its components in the three cartesian coordinates of the LV reference system, which could then be transformed in the components parallel and perpendicular to an arbitrary surface, as shown in Fig. 5, where the field on one of the faces of the envelope directly above the antennas was

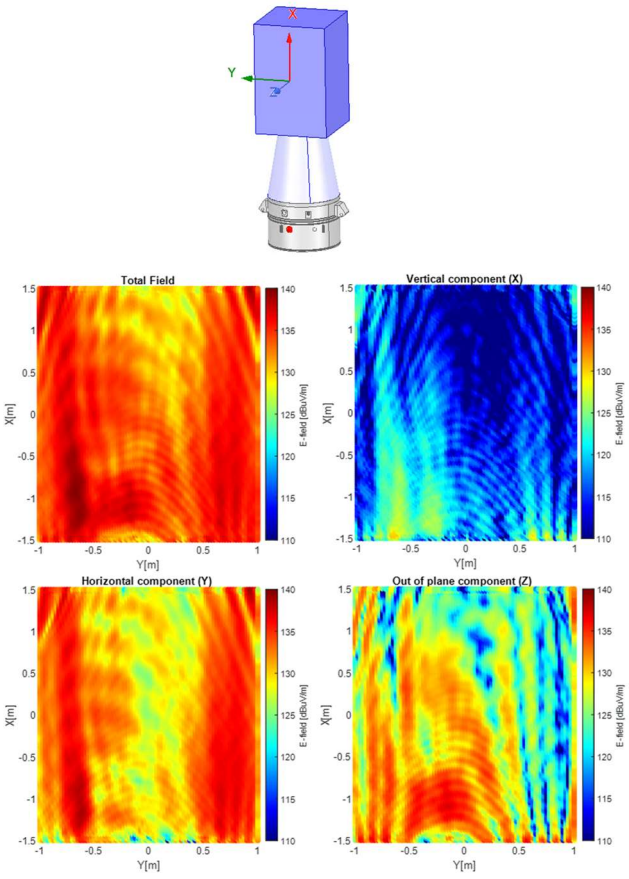


Fig. 5 Decomposition of E-field in components parallel and perpendicular to a face of the envelope (0.1 seconds after fairing separation). As expected, the vertical component (along X-axis) is very small, as the direction of propagation of the EM wave is mostly parallel to the X axis.

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decomposed (the time instant considered in this example was the worst case at t = 0.1 seconds). It is interesting to notice that the X component of the field was significantly smaller than the other two components. This was expected, as the source antenna was directly below the panel, and therefore the EM wave propagation direction was almost parallel to the X axis.

This feature can be particularly useful to analyze payloads that are susceptible to specific polarizations, as could be the case for satellites having arrays of linearly polarized elements.

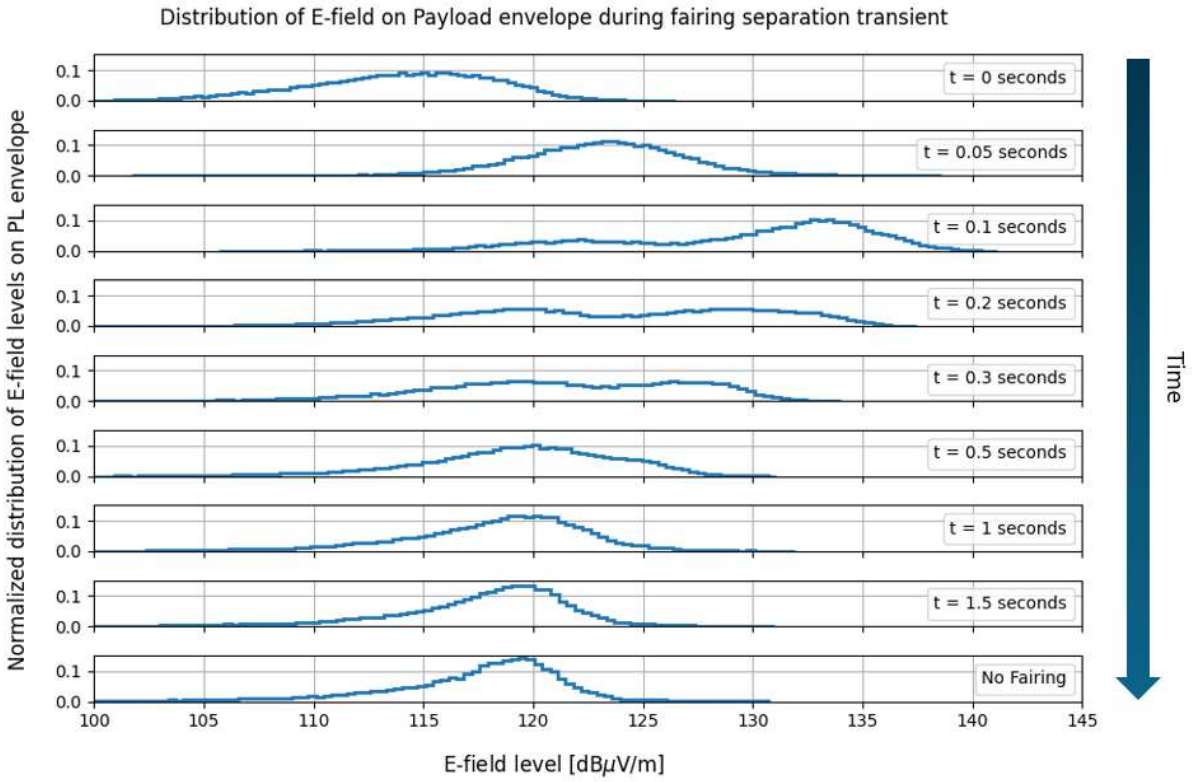


Fig. 6 Normalized distribution of E-field values on payload envelope surface at different instants of the separation transient. Highest values were reached after 0.1 seconds, while at 1 second the effect of the fairing was almost negligible.

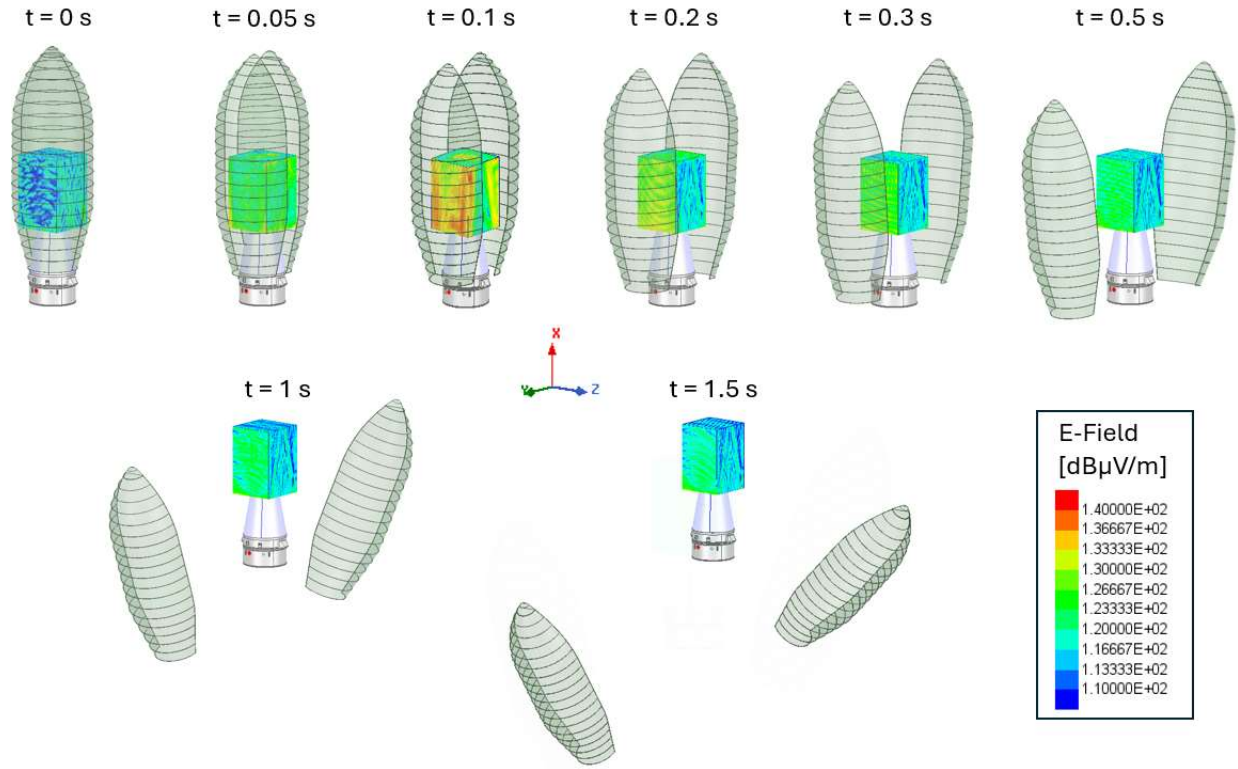


Fig. 7 Simulation of the fairing separation sequence with E-field levels on fairing envelope. It can be observed that as expected the effect of the fairing is more prominent on the face that is directly in front of one of the shells.

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V. CONCLUSIONS

In conclusion, in this paper we presented the model that was developed to study the E-field caused by two antennas of a Launch Vehicle on a generic payload surface during the LV fairing separation transient. The model was shown to return realistic results, with less than 3 dB of overestimation that were accepted as additional margin. In particular, it was observed that the E-field on a generic payload envelope increased until about 0.1 seconds after the fairing separation, then started to decrease until it reached a steady state after about 1 second, confirming that the effects of the fairing reflection are significant for a very brief transient.

However, there is still room for improving the model: in particular, future efforts will focus on substituting the imported antenna measurements with a representative finite element model of the antenna, developed directly in ANSYS HFSS. This will improve the near-field interactions of the antenna with the surrounding structures. Furthermore, the characteristics of each material will be implemented instead of the PEC structures. This will allow to obtain better accuracy in the estimation of reflection and to also study the impact of the addition of specific materials that could mitigate the effect, such as absorbing foams placed inside the fairing shells.

Finally, this model can be used for studies on both existing launchers and new launchers under development and is part of an effort by Avio to implement and improve models in the EMC analyses of launch vehicles.

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